

CS1675/2075: Introduction to Machine Learning

Week 7: Linear Models Deep Dive

Fall 2024

“Fitting” a model?

- We collect data consisting of INPUTS and OUTPUTS.
- For example, if there is a single INPUT and a single OUTPUT then we will collect N input-output pairs.
- Fitting the model refers to learning or estimating the model's unknown parameters.
- What are the unknowns?

Let's continue to use a linear relationship between the mean trend and the input.

- Start out assuming that σ is known.
- We need to learn the unknown **INTERCEPT** and the unknown **SLOPE**

$$\mu_n = \beta_0 + \beta_1 x_n$$

- How can we do that? What values would be “best”?

Maximize the likelihood!

$$p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma) = \prod_{n=1}^N \{p(y_n \mid \mu_n, \sigma)\} = \prod_{n=1}^N \{\text{normal}(y_n \mid \mu_n, \sigma)\}$$

$$\mu_n = \beta_0 + \beta_1 x_n$$

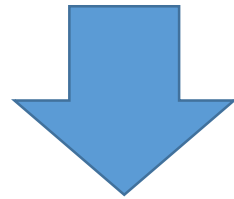
- The learning problem is therefore an optimization problem!

$$\boldsymbol{\beta}_{MLE} = \max p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)$$

First, consider the log-likelihood in terms of the mean trend μ_n

- The log-likelihood is the sum of N conditionally independent Gaussian log-likelihoods:

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] = \sum_{n=1}^N \{\log[\text{normal}(y_n \mid \mu_n, \sigma)]\}$$



$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] = \sum_{n=1}^N \left\{ -\frac{1}{2} \log[\sigma^2] - \frac{1}{2} \log[2\pi] - \frac{1}{2\sigma^2} (y_n - \mu_n)^2 \right\}$$

Drop the constants

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] \propto -\frac{1}{2\sigma^2} \sum_{n=1}^N \{(y_n - \mu_n)^2\}$$

Substitute in the expression for the mean trend

$$y_n - \mu_n = y_n - (\beta_0 + \beta_1 x_n) = \epsilon_n$$

The n -th observation's error or residual!!

The summation term is the **residual sum of squares (RSS)** or the **sum of squared errors (SSE)**!

$$RSS = SSE = \sum_{n=1}^N \{(y_n - \mu_n)^2\} = \sum_{n=1}^N \{\epsilon_n^2\}$$

- The log-likelihood is therefore proportional to the negative of the SSE:

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] \propto -SSE$$

The summation term is the **residual sum of squares (RSS)** or the **sum of squared errors (SSE)**!

$$\log[p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)] \propto -SSE$$

- Maximizing the (log) likelihood is therefore equivalent to minimizing the squared error!

The Maximum Likelihood Estimates (MLEs) to the linear model parameters will be equivalent to the Ordinary Least Squares (OLS) estimates!

Bayesian formulation

- Still consider σ to be known.
- We need to specify our prior belief about the unknown β 's.
- We update our prior based on the observations, to yield the posterior on the unknowns.

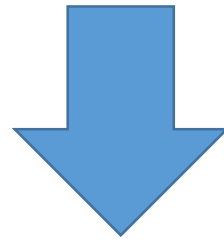
Bayesian formulation

$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{x}, \sigma) \propto p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma) \times p(\boldsymbol{\beta})$$

- If we would use a diffuse, uninformative, or ignorant prior on $\boldsymbol{\beta}$: $p(\boldsymbol{\beta}) \propto 1...$
- How does the posterior behave?

Bayesian formulation with a very diffuse or uninformative prior follows the likelihood!

$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{x}, \sigma) \propto p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma) \times 1$$



$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{x}, \sigma) \propto p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)$$

We know that the MLE is equivalent to the OLS estimate.
We should expect that with a diffuse prior the MAP should also correspond to the OLS estimate!

Bayesian formulation with an informative prior

$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{x}, \sigma) \propto p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma) \times p(\boldsymbol{\beta})$$

- The likelihood is still the same...it wants to minimize the error...
- What impact will the prior have?

We usually start by visualizing the results, but let's focus on the math first

- We will begin by assuming σ is known and then we will relax that assumption later.
- We will also assume a very diffuse prior on $\boldsymbol{\beta}$.
- As we saw previously, the posterior then follows the likelihood:

$$p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{x}, \sigma) \propto p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\beta}, \sigma)$$

Since σ is known, the log-posterior simplifies to:

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{x}, \sigma)] \propto -\frac{1}{2\sigma^2} \sum_{n=1}^N \{(y_n - \mu_n)^2\}$$

- We previously discussed how the summation term is just the RSS or SSE.

We have focused on a simple case of a linear relationship with a single input

$$\mu_n = \beta_0 + \beta_1 x_n$$

- We can work through the mathematical details for the estimates and uncertainty on the unknown INTERCEPT, β_0 , and SLOPE, β_1 .
- However, we want an approach that will scale to many coefficients!
- We want to generalize the procedure for learning dozens if not hundreds of unknown regression coefficients!

Let's try and generalize our formulation to handle more than one input

- For example if we had 3 inputs, $\{x_1, x_2, x_3\}$, the linear model with linear relationships would be:

$$\mu_n = \beta_0 + \beta_1 x_{n,1} + \beta_2 x_{n,2} + \beta_3 x_{n,3}$$

- There are now MULTIPLE slopes because we have MULTIPLE inputs!

Please note the change in subscript notation between the SINGLE input case and the MULTIPLE input case

- ONE subscript is used for the SINGLE input

$$\mu_n = \beta_0 + \beta_1 x_n$$

- However, TWO subscript indices are used when we have MULTIPLE inputs.
- For example if we had 3 inputs, $\{x_1, x_2, x_3\}$, the linear model with linear relationships would be:

$$\mu_n = \beta_0 + \beta_1 x_{n,1} + \beta_2 x_{n,2} + \beta_3 x_{n,3}$$

Subscript denotes the n -th observation of input 1

Let's revisit the expression for the mean trend

- The trend includes new *features* by multiplying the *feature* by a coefficient.

$$\mu_n = \beta_0 + \beta_1 x_{n,1} + \beta_2 x_{n,2} + \beta_3 x_{n,3}$$

- Write the multiplication in GENERAL terms as the product of the d -th feature, $x_{n,d}$, and its corresponding d -th slope, β_d :

$$\beta_d x_{n,d}$$

Sum the PRODUCTS across the 3 inputs!

$$\mu_n = \beta_0 + \beta_1 x_{n,1} + \beta_2 x_{n,2} + \beta_3 x_{n,3} = \beta_0 + \sum_{d=1}^3 (\beta_d x_{n,d})$$

Generalize to handle an arbitrary number of INPUTS!! Denote the number of features as D

$$\mu_n = \beta_0 + \beta_1 x_{n,1} + \cdots + \beta_d x_{n,d} + \cdots + \beta_D x_{n,D}$$



$$\mu_n = \beta_0 + \sum_{d=1}^D (\beta_d x_{n,d})$$

Can rewrite the mean trend with a summation expression over all D inputs

$$\mu_n = \beta_0 + \sum_{d=1}^D (\beta_d x_{n,d})$$

- How can we put the intercept within the summation?

Let's create a “fake” variable, $x_{n,0}$, which is defined to always equal 1.

- With a single input, we will rewrite the mean trend with the fake input $x_{n,0}$ and denote the actual input as $x_{n,1}$:

$$\mu_n = \beta_0 x_{n,0} + \beta_1 x_{n,1}$$

- The summation over D inputs then becomes:

$$\mu_n = \beta_0 x_{n,0} + \sum_{d=1}^D (\beta_d x_{n,d})$$

Bring the intercept term into the summation

$$\mu_n = \sum_{d=0}^D (\beta_d x_{n,d})$$


Summation starts at ZERO not ONE

- Now, the summation acts over $D + 1$ terms.
- $\beta_{d=0}$ is just the intercept.

The notation $x_{n,d}$ corresponds to the n -th observation of the d -th input

- We can therefore think of the **observation label**, n , as the row index and the **input label**, d , as a column index.
- Lends itself nicely to organizing all of the inputs together in a matrix!
- The matrix will consist of N rows, one for each observation, and $D + 1$ columns, one for each input plus the intercept.

This matrix of inputs will be referred to as the DESIGN MATRIX

- With a single input and single linear feature the design matrix is:

$$\mathbf{X} = \begin{bmatrix} x_{1,0} & x_{1,1} \\ \vdots & \vdots \\ x_{N,0} & x_{N,1} \end{bmatrix} = \begin{bmatrix} 1 & x_{1,1} \\ \vdots & \vdots \\ 1 & x_{N,1} \end{bmatrix}$$

In general, with D linear additive inputs the DESIGN MATRIX is:

$$\mathbf{X} = \begin{bmatrix} x_{1,0} & x_{1,1} & \cdots & x_{1,D} \\ x_{2,0} & x_{2,1} & \cdots & x_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,0} & x_{N,1} & \cdots & x_{N,D} \end{bmatrix} = \begin{bmatrix} 1 & x_{1,1} & \cdots & x_{1,D} \\ 1 & x_{2,1} & \cdots & x_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N,1} & \cdots & x_{N,D} \end{bmatrix}$$

 The 1st input column
 $\mathbf{x}_{:,1}$

 The D -th input column
 $\mathbf{x}_{:,D}$

We can now rewrite the mean trend using vector-matrix notation

- The n -th row of the design matrix is a $1 \times (D + 1)$ row vector:

$$\mathbf{X}_{n,:}$$

- For the single input case, the n -th row is:

$$\mathbf{x}_{n,:} = [1 \quad x_{n,1}]$$

- For three inputs:

$$\mathbf{x}_{n,:} = [1 \quad x_{n,1} \quad x_{n,2} \quad x_{n,3}]$$

We can now rewrite the mean trend using vector-matrix notation

- The n -th row of the design matrix is a $1 \times (D + 1)$ row vector:

$$\mathbf{X}_{n,:}$$

- Organize the $\boldsymbol{\beta}$ coefficients into a $(D + 1) \times 1$ column vector.
- For a single input:

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

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$$\mathbf{X}_{n,:}$$

- Organize the $\boldsymbol{\beta}$ coefficients into a $(D + 1) \times 1$ column vector.
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Rewrite the mean trend using vector-matrix notation

- The n -th row of the design matrix is a $1 \times (D + 1)$ row vector:

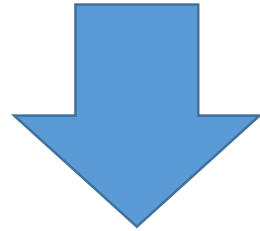
$$\mathbf{x}_{n,:}$$

- Organize the $\boldsymbol{\beta}$ coefficients into a $(D + 1) \times 1$ column vector.
- The mean trend associated with the n -th observation can then be written as the inner-product of the row and column vectors:

$$\mu_n = \sum_{d=0}^D (\beta_d x_{n,d}) = \mathbf{x}_{n,:} \boldsymbol{\beta}$$

For the single input case...

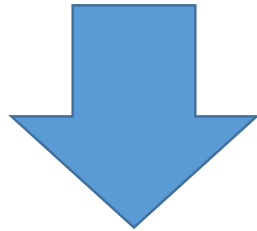
$$\mu_n = \mathbf{x}_{n,:} \boldsymbol{\beta} = [1 \quad x_{n,1}] \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$



$$\mu_n = \beta_0 \times 1 + \beta_1 \times x_{n,1} = \beta_0 + \beta_1 x_{n,1}$$

Same idea for multiple inputs!

$$\mu_n = \mathbf{x}_{n,:} \boldsymbol{\beta} = [1 \quad x_{n,1} \quad x_{n,2} \quad x_{n,3}] \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}$$



$$\mu_n = \beta_0 \times 1 + \beta_1 \times x_{n,1} + \beta_2 \times x_{n,2} + \beta_3 \times x_{n,3}$$

$$\mu_n = \beta_0 + \beta_1 x_{n,1} + \beta_2 x_{n,2} + \beta_3 x_{n,3}$$

The vector of mean trends for all N observations corresponds to “stacking” the inner products together

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_{n=1} \\ \mu_{n=2} \\ \vdots \\ \mu_{n=N} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_{1,:} \boldsymbol{\beta} \\ \mathbf{x}_{2,:} \boldsymbol{\beta} \\ \vdots \\ \mathbf{x}_{N,:} \boldsymbol{\beta} \end{bmatrix}$$

We can perform all calculations at once with the matrix multiplication of the design matrix and coefficient vector:

$$\mu = X\beta$$

The summation term in the log-posterior can then be rewritten using the matrix-vector notation

$$-\frac{1}{2\sigma^2} \sum_{n=1}^N \{(y_n - \mu_n)^2\} = -\frac{1}{2\sigma^2} \sum_{n=1}^N \{(y_n - \mathbf{x}_{n,:}\boldsymbol{\beta})^2\}$$



$$-\frac{1}{2\sigma^2} \sum_{n=1}^N \{(y_n - \mu_n)^2\} = -\frac{1}{2\sigma^2} \underbrace{(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})}_{SSE}$$

The log-likelihood written to handle an arbitrary number of inputs and thus an arbitrary number of unknown regression coefficients:

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto -\frac{1}{2\sigma^2} \sum_{n=1}^N \{(y_n - \mu_n)^2\}$$

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto -\frac{1}{2\sigma^2} \sum_{n=1}^N \{(y_n - \mathbf{x}_{n,:}\boldsymbol{\beta})^2\}$$

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto -\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto -\frac{1}{2\sigma^2} SSE$$

Why do we need the log-likelihood?

- In Non-Bayesian settings, we must MAXIMIZE the log-likelihood to ESTIMATE the unknown regression coefficients.
- In Bayesian settings, the log-likelihood is a component of the log-posterior!
 - If the Prior does not matter, the log-posterior follows the log-likelihood!
- If we use the Laplace Approximation as our Bayesian inference engine...then we must also MAXIMIZE a function!

The Laplace Approximation requires:

- We need to find the posterior mode, or the MAP, of the unknowns.
 - OPTIMIZATION -> requires the gradient
- Then, we will approximate the posterior covariance matrix as the negative inverse of the Hessian matrix.
 - Requires the second derivatives!

Denote the log-likelihood as L

$$L = -\frac{1}{2\sigma^2} \sum_{n=1}^N \left\{ (y_n - \mathbf{x}_{n,:}\boldsymbol{\beta})^2 \right\} = -\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

- We need to determine the partial first derivatives of L with respect to each element of $\boldsymbol{\beta}$.
- Remember we are assuming the prior does not matter and so the log-posterior is proportional to the log-likelihood!

The complete derivation that goes through all steps to calculate $\partial L / \partial \beta_d$ are provided on Canvas

- For now, let's take a short cut...

$$L = -\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

The inner product or summation of
squared elements of a vector

The complete derivation that goes through all steps to calculate $\partial L / \partial \beta_d$ are provided on Canvas

- For now, let's take a short cut...
- $L = -\underbrace{\frac{1}{2\sigma^2}}_{\text{constant multiplier}} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ (As a reminder)

The constant “multiplier” out front.

$$\mathbf{g} = \begin{bmatrix} \frac{\partial L}{\partial \beta_{d=0}} \\ \frac{\partial L}{\partial \beta_{d=1}} \\ \vdots \\ \frac{\partial L}{\partial \beta_{d=D}} \end{bmatrix} = -\frac{1}{2\sigma^2}$$

The complete derivation that goes through all steps to calculate $\partial L / \partial \beta_d$ are provided on Canvas

- For now, let's take a short cut...
- $L = -\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ (As a reminder)

Matrix notation for “raised to the second power”...bring down the exponent...

$$\mathbf{g} = \begin{bmatrix} \frac{\partial L}{\partial \beta_{d=0}} \\ \frac{\partial L}{\partial \beta_{d=1}} \\ \vdots \\ \frac{\partial L}{\partial \beta_{d=D}} \end{bmatrix} = -\frac{1}{2\sigma^2} \times 2$$

The complete derivation that goes through all steps to calculate $\partial L / \partial \beta_d$ are provided on Canvas

- For now, let's take a short cut...
- $L = -\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ (As a reminder)

Derivative “inside the parentheses”

$$\mathbf{g} = \begin{bmatrix} \frac{\partial L}{\partial \beta_{d=0}} \\ \frac{\partial L}{\partial \beta_{d=1}} \\ \vdots \\ \frac{\partial L}{\partial \beta_{d=D}} \end{bmatrix} = -\frac{1}{2\sigma^2} \times 2 \times -\mathbf{X}^T$$

The complete derivation that goes through all steps to calculate $\partial L / \partial \beta_d$ are provided on Canvas

- For now, let's take a short cut...

- $L = -\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ (As a reminder)

Rewrite the term in the parentheses

$$\mathbf{g} = \begin{bmatrix} \frac{\partial L}{\partial \beta_{d=0}} \\ \frac{\partial L}{\partial \beta_{d=1}} \\ \vdots \\ \frac{\partial L}{\partial \beta_{d=D}} \end{bmatrix} = -\frac{1}{2\sigma^2} \times 2 \times -\mathbf{X}^T \times (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

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$$\mathbf{g} = \begin{bmatrix} \frac{\partial L}{\partial \beta_{d=0}} \\ \frac{\partial L}{\partial \beta_{d=1}} \\ \vdots \\ \frac{\partial L}{\partial \beta_{d=D}} \end{bmatrix} = -\frac{1}{2\sigma^2} \times 2 \times -\mathbf{X}^T \times (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

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- For now, let's take a short cut...

$$\mathbf{g} = \begin{bmatrix} \frac{\partial L}{\partial \beta_{d=0}} \\ \frac{\partial L}{\partial \beta_{d=1}} \\ \vdots \\ \frac{\partial L}{\partial \beta_{d=D}} \end{bmatrix} = \frac{1}{\sigma^2} (\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta})$$

The gradient vector:

$$\mathbf{g} = \frac{1}{\sigma^2} (\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta})$$

Two key terms:

The projection of the RESPONSE onto
the INPUT space



$$\mathbf{g} = \frac{1}{\sigma^2} (\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta})$$

Two key terms:

The relationships between
the inputs!!!


$$\mathbf{g} = \frac{1}{\sigma^2} (\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta})$$

Sum of squares matrix:

$$\sum_{n=1}^N (\mathbf{x}_{n,:}^T \mathbf{x}_{n,:}) = \mathbf{X}^T \mathbf{X}$$

If the inputs have been centered...the sum of squares essentially represents the covariance of the inputs!

Why is it called sum of squares? Because the inputs are squared and then summed over the observations!

$$\sum_{n=1}^N \left(\mathbf{X}_{n,:}^T \mathbf{X}_{n,:} \right) = \sum_{n=1}^N \begin{bmatrix} x_{n,0}^2 & x_{n,0}x_{n,1} & \cdots & x_{n,0}x_{n,D} \\ x_{n,1}x_{n,0} & x_{n,1}^2 & \cdots & x_{n,1}x_{n,D} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,D}x_{n,0} & x_{n,D}x_{n,1} & \cdots & x_{n,D}^2 \end{bmatrix}$$

Sum of squares matrix is $(D + 1) \times (D + 1)$

The mode is found by setting gradient vector equal to 0, $\mathbf{g} = \mathbf{0}$

- Denote the mode as $\hat{\boldsymbol{\beta}}$:

$$0 = \frac{1}{\sigma^2} (\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}})$$

- Rearrange and solve for $\hat{\boldsymbol{\beta}}$:

$$\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{y}$$

Since we started from a very diffuse prior...

The posterior mode is just the MLEs which equals the OLS estimators on the coefficients!

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Since we started from a very diffuse prior...

The posterior mode is just the MLEs which equals the OLS estimators on the coefficients!

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

The noise, σ , does **NOT** explicitly impact the MLEs!!

Since we started from a very diffuse prior...

The posterior mode is just the MLEs which equals the OLS estimators on the coefficients!

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Need to be able to invert the sum of squares matrix!!

What if we have an INTERCEPT-ONLY model...

- An intercept-only model ONLY consists of the INTERCEPT:

$$\mu_n = \beta_0 x_{n,0} = \beta_0 \times 1 = \beta_0$$

The design matrix has a SINGLE column:

$$\mathbf{X} = \begin{bmatrix} x_{1,0} \\ \vdots \\ x_{N,0} \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

What if we have an INTERCEPT-ONLY model...

- Calculate the MLE for the single unknown INTERCEPT:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \left(\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^T \mathbf{y}$$

What if we have an INTERCEPT-ONLY model...

- Calculate the MLE for the single unknown INTERCEPT:

$$\hat{\boldsymbol{\beta}} = \left(\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^T \mathbf{y} = \left([1 \quad \dots \quad 1] \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^T \mathbf{y}$$

What if we have an INTERCEPT-ONLY model...

- Calculate the MLE for the single unknown INTERCEPT:

$$\hat{\boldsymbol{\beta}} = \underbrace{\begin{pmatrix} [1 & \dots & 1] & \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \end{pmatrix}}_{\text{INNER PRODUCT!!!!}}^{-1} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^T \mathbf{y}$$

$$= \sum_{n=1}^N (1_n \times 1_n) = \sum_{n=1}^N (1) = N$$

What if we have an INTERCEPT-ONLY model...

- Calculate the MLE for the single unknown INTERCEPT:

$$\hat{\boldsymbol{\beta}} = \left(\begin{bmatrix} 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right)^{-1} \underbrace{\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^T \mathbf{y}}_{\text{INNER PRODUCT!!!!}}$$
$$= \sum_{n=1}^N (1_n \times y_n) = \sum_{n=1}^N (y_n) = N\bar{y}$$

What if we have an INTERCEPT-ONLY model...

- Calculate the MLE for the single unknown INTERCEPT:

$$\hat{\beta} = (N)^{-1} \times N\bar{y} = \bar{y}$$

- The INTERCEPT-ONLY model is the unknown CONSTANT average!
- The unknown constant average is the simplest linear model!!!!

With the mode identified, now we need to evaluate the Hessian

- The Hessian matrix is the matrix of second derivatives.
- For the single input case and a very diffuse prior, the Hessian matrix looks like:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 L}{\partial \beta_0^2} & \frac{\partial^2 L}{\partial \beta_0 \partial \beta_1} \\ \frac{\partial^2 L}{\partial \beta_1 \partial \beta_0} & \frac{\partial^2 L}{\partial \beta_1^2} \end{bmatrix}$$

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- The Hessian matrix is the matrix of second derivatives.
- For the 3 inputs and a very diffuse prior, the Hessian matrix looks like:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 L}{\partial \beta_0^2} & \frac{\partial^2 L}{\partial \beta_0 \partial \beta_1} & \frac{\partial^2 L}{\partial \beta_0 \partial \beta_2} & \frac{\partial^2 L}{\partial \beta_0 \partial \beta_3} \\ \frac{\partial^2 L}{\partial \beta_1 \partial \beta_0} & \frac{\partial^2 L}{\partial \beta_1^2} & \frac{\partial^2 L}{\partial \beta_1 \partial \beta_2} & \frac{\partial^2 L}{\partial \beta_1 \partial \beta_3} \\ \frac{\partial^2 L}{\partial \beta_2 \partial \beta_0} & \frac{\partial^2 L}{\partial \beta_2 \partial \beta_1} & \frac{\partial^2 L}{\partial \beta_2^2} & \frac{\partial^2 L}{\partial \beta_2 \partial \beta_3} \\ \frac{\partial^2 L}{\partial \beta_3 \partial \beta_0} & \frac{\partial^2 L}{\partial \beta_3 \partial \beta_1} & \frac{\partial^2 L}{\partial \beta_3 \partial \beta_2} & \frac{\partial^2 L}{\partial \beta_3^2} \end{bmatrix}$$

Rather than going through all steps to determine the Hessian...

- Let's just take a look at the gradient vector...which part of the gradient depends on the coefficients?

$$\mathbf{g} = \frac{1}{\sigma^2} (\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta})$$



$$\mathbf{H} = -\frac{1}{\sigma^2} \mathbf{X}^T \mathbf{X}$$

We previously saw that the regression coefficient MLEs have a closed form analytic solution!

- However, let's pretend we forgot that.
- Let's pretend we felt we needed an optimization algorithm to find the MLEs.
- We need an initial guess, $\beta_{k=0}$, and we will use an iterative optimization algorithm to find the optimal coefficients that maximize the likelihood.

We have formulas for the gradient and Hessian...so let's use the Newton Method!

$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} - \gamma_{k=0} \mathbf{H}_{k=0}^{-1} \mathbf{g}_{k=0}$$

- Let's use a FULL Newton step, $\gamma_{k=0} = 1$, the update formula becomes:

$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} - \mathbf{H}_{k=0}^{-1} \mathbf{g}_{k=0}$$

Substitute the expressions for the gradient and Hessian! Simplify the algebra!!!

$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} - \left(-\frac{1}{\sigma^2} \mathbf{X}^T \mathbf{X} \right)^{-1} \left(\frac{1}{\sigma^2} (\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}_{k=0}) \right)$$



$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} \left(\frac{1}{\sigma^2} (\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}_{k=0}) \right)$$



$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} ((\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}_{k=0}))$$

Substitute the expressions for the gradient and Hessian! Simplify the algebra!!!

$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} ((\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}_{k=0}))$$



$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} - (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}_{k=0}$$

Substitute the expressions for the gradient and Hessian! Simplify the algebra!!!

$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} ((\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}_{k=0}))$$



$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} - \boxed{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}_{k=0}}$$



A matrix multiplied by its inverse!!!!
Equals the IDENTITY matrix!!!!

$$(\mathbf{X}^T \mathbf{X})^{-1} (\mathbf{X}^T \mathbf{X}) = \mathbb{I} = \begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix}$$

Substitute the expressions for the gradient and Hessian! Simplify the algebra!!!

$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} ((\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}_{k=0}))$$



$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} - (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}_{k=0}$$



$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} - \mathbb{I} \boldsymbol{\beta}_{k=0}$$

Substitute the expressions for the gradient and Hessian! Simplify the algebra!!!

$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} ((\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}_{k=0}))$$



$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} - (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}_{k=0}$$



$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} - \mathbb{I} \boldsymbol{\beta}_{k=0}$$



$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} - \boldsymbol{\beta}_{k=0}$$

The Newton method converges in 1 iteration!

$$\boldsymbol{\beta}_{k=1} = \boldsymbol{\beta}_{k=0} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} - \boldsymbol{\beta}_{k=0}$$



$$\boldsymbol{\beta}_{k=1} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Why does this matter?

- The Hessian matrix does **NOT** depend on the regression coefficients!!!!!!!!!!!!!!

$$\mathbf{H} = -\frac{1}{\sigma^2} \mathbf{X}^T \mathbf{X}$$

- The second derivatives are constants and thus the **UNKNOWN** regression coefficients are **LINEARLY** related to the log-likelihood!!!

If we're using the Laplace Approximation to perform the Bayesian inference...

- We need to evaluate the Hessian matrix at the posterior mode.
- The approximate covariance matrix is the negative inverse of the Hessian at the posterior mode!

$$\text{cov}(\boldsymbol{\beta}, \boldsymbol{\beta}) = [-\mathbf{H}_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}}]^{-1} = \left(-1 \times -\frac{1}{\sigma^2} \mathbf{X}^T \mathbf{X} \right)^{-1} = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

The approximate posterior covariance matrix is the negative inverse of the Hessian:

$$\text{cov}(\boldsymbol{\beta}, \boldsymbol{\beta}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

- The covariance between the coefficients does NOT depend on the response!

The approximate posterior covariance matrix is the negative inverse of the Hessian:

$$\text{cov}(\boldsymbol{\beta}, \boldsymbol{\beta}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

- The covariance between the coefficients does NOT depend on the response!
- Need to be able to invert the sum of squares matrix!

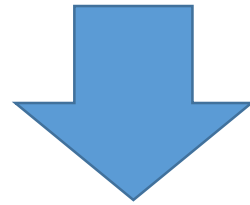
The approximate posterior covariance matrix is the negative inverse of the Hessian:

$$\text{cov}(\boldsymbol{\beta}, \boldsymbol{\beta}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

- The covariance between the coefficients does NOT depend on the response!
- Need to be able to invert the sum of squares matrix!
- The input covariance therefore impacts the covariance of the coefficients!

The MVN posterior distribution based on a very diffuse or uninformative prior is:

$$\boldsymbol{\beta} \mid \mathbf{X}, \mathbf{y}, \sigma \sim \mathcal{N}(\boldsymbol{\beta} \mid \hat{\boldsymbol{\beta}}, \text{cov}(\boldsymbol{\beta}, \boldsymbol{\beta}))$$



$$\boldsymbol{\beta} \mid \mathbf{X}, \mathbf{y}, \sigma \sim \mathcal{N}(\boldsymbol{\beta} \mid (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1})$$

What if we have an **INTERCEPT-ONLY** model...

- The Hessian Matrix for the log-posterior involves the sum of squares of a design matrix with a single column!

$$\mathbf{H} = -\frac{1}{\sigma^2} \mathbf{X}^T \mathbf{X} = -\frac{1}{\sigma^2} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = ?$$

What if we have an **INTERCEPT-ONLY** model...

- The Hessian Matrix for the log-posterior involves the sum of squares of a design matrix with a single column!

$$\mathbf{H} = -\frac{1}{\sigma^2} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = -\frac{1}{\sigma^2} [1 \quad \dots \quad 1] \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

What if we have an **INTERCEPT-ONLY** model...

- The Hessian Matrix for the log-posterior involves the sum of squares of a design matrix with a single column!

$$\mathbf{H} = -\frac{1}{\sigma^2} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = -\frac{1}{\sigma^2} \underbrace{\begin{bmatrix} 1 & \dots & 1 \end{bmatrix}}_{\sum_{n=1}^N (1_n \times 1_n) = \sum_{n=1}^N (1) = N} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

What if we have an **INTERCEPT-ONLY** model...

- The Hessian Matrix for the log-posterior involves the sum of squares of a design matrix with a single column!

$$\mathbf{H} = -\frac{1}{\sigma^2} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = -\frac{1}{\sigma^2} \times N = -\frac{N}{\sigma^2}$$

What if we have an **INTERCEPT-ONLY** model...

- The negative inverse of the Hessian matrix is the posterior covariance matrix:

$$\text{cov}(\boldsymbol{\beta}, \boldsymbol{\beta}) = [-\mathbf{H}]^{-1} = \left(-1 \times -\frac{N}{\sigma^2} \right)^{-1} = \frac{\sigma^2}{N}$$

What if we have an **INTERCEPT-ONLY** model...

- The posterior distribution on the unknown intercept is a Gaussian with mean equal to the **SAMPLE AVERAGE** and standard deviation equal to the standard error formula!!!!

$$\boldsymbol{\beta} \mid \mathbf{X}, \mathbf{y}, \sigma \sim \mathcal{N}(\boldsymbol{\beta} \mid (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1})$$



$$\beta_0 \mid \mathbf{X}, \mathbf{y}, \sigma \sim \text{normal}\left(\beta_0 \mid \bar{\mathbf{y}}, \frac{\sigma}{\sqrt{N}}\right)$$

This is the normal-normal model!!!!!!!!!!!!!!